

Exercise 7: Set and Parameter Actions

Three dashboards, three captions. No memo.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 11. Tableau Help: Set Actions (help.tableau.com/current/pro/desktop/en-us/actions_sets.htm) and Parameter Actions (help.tableau.com/current/pro/desktop/en-us/actions_parameters.htm)
Walkthrough	Module 7 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with three dashboards (Worksheets 1–3) and a caption on each. No memo.

Set up

1. Connect mvu_student_dim.csv. Set student_id to a Dimension.
2. Create Graduation Rate = $\text{AVG}([\text{is_graduated}])$ and Withdrawal Rate = $\text{AVG}([\text{is_withdrawn}])$, both formatted as Percentage, 1 decimal.
3. On every sheet that shows a rate, filter cohort_year to 2020 and 2021.

Checkpoint: Graduation rate for the 2020 and 2021 cohorts combined?

Worksheet 1: let the reader build the comparison

1. Create a set on program_name named Comparison Programs, with Data Science ticked.
2. Sheet Programs: program_name on Rows, student_id as Count (Distinct) on Columns, sorted descending, the set on Color.
3. Sheet In vs Out: the set on Rows, Graduation Rate and Withdrawal Rate on Columns, labels on, 2020–21 cohorts.

Checkpoint: With only Data Science in the set, what is the IN graduation rate?

4. Dashboard Worksheet 1 with both sheets. Add a Change Set Values action: source Programs; run on Select; target Comparison Programs; Add values to set; clearing keeps set values.
5. Add a second action from Programs, run on Menu, that removes values from the set.
6. Click Public Health and Accounting. Then remove the cohort filter from In vs Out for a moment and compare; put it back.

Checkpoint: IN graduation rate (2020–21) with Data Science, Public Health and Accounting in the set?

Worksheet 2: a parameter action sets the benchmark

1. Create a Float parameter Benchmark Rate (current value 0.772, percentage format).
2. Create Vs Benchmark: IF [Graduation Rate] >= [Benchmark Rate] THEN "At or above benchmark" ELSE "Below benchmark" END.

3. Sheet: program_name on Rows, Graduation Rate on Columns, sorted descending, Vs Benchmark on Color, 2020–21 cohorts, and a reference line at Benchmark Rate. Title it with the parameter value.
4. Dashboard Worksheet 2. Add a Change Parameter action: source the bar sheet; target Benchmark Rate; source field Graduation Rate; run on Select; clearing keeps the current value.

Checkpoint: With Business Administration clicked as the benchmark, how many programs are at or above it?

Worksheet 3: proportional brushing

1. Create a second set on program_name, Selected Programs, with no members.
2. Scatter: Graduation Rate on Columns, Withdrawal Rate on Rows, program_name on Detail, 2020–21 cohorts.
3. Bars: campus_name on Rows, student_id as Count (Distinct), Selected Programs on Color, all cohorts. Label with Share in Selection = $\text{SUM}(\text{IIF}([\text{Selected Programs}], 1, 0)) / \text{COUNT}([\text{student_id}])$.
4. Dashboard Worksheet 3. Change Set Values action: source the scatter; target Selected Programs; run on Select; Assign values to set; clearing removes all values.

Checkpoint: With Computer Science and Data Science selected, what share of Riverside Campus students are in the selection?

Your three captions

Put each caption in a Text object below the views on its dashboard: one or two sentences answering one question.

Dashboard	What your caption must answer
Worksheet 1	Build a set of your choice. How do its programs compare with everyone else, and why does the answer change when all cohorts are included?
Worksheet 2	What does clicking a bar do, and how would a first-time reader know they can? When is a program-as-benchmark more useful than a fixed target?
Worksheet 3	What does the brushed share tell a reader that the scatter alone can't? What should they do first, and what does 0% mean?

Worksheet 1's caption carries 25% on its own. Reporting the IN and OUT rates without the cohort point does not earn full marks. Interactivity lets the reader build any group; your job is to keep the comparison fair.

Keep it fast: keep action sources at program level, not student level, and test each action. If a click takes more than a second to redraw, look for marks you don't need (Help > Settings and Performance > Start Performance Recording).

Submit for Exercise 7

- Worksheet 1: a dashboard where clicking programs adds them to a set, with an IN-versus-OUT comparison on 2020–21 cohorts and a way to remove programs
- Worksheet 2: a dashboard where clicking a program sets the benchmark through a parameter action, with a reference line and recoloring
- Worksheet 3: a proportional-brushing dashboard where a program scatter drives a set that splits campus headcount

- A caption on each
- Packaged workbook (.twbx). No memo.

Optional extensions

- Click a header to sort: a Sort Metric parameter and a Dynamic Sort Measure (as in Exercise 4), changed by a parameter action from a small header sheet. Test it in your version of Tableau first.
- Asymmetric drill-down: a set on college, a field IF [Expanded Colleges] THEN [program_name] ELSE "" END, and a set action that opens only the clicked college.